

Abdullah Al-Nafisah

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Education

Master of Science in Electrical and Computer Engineering

2022 - 2024

King Abdullah University of Science and Technology (KAUST), Thuwal, KSA

GPA: 3.83/4.0

Thesis: *Turbo-Equalized FTN-Inspired Modulations*, doi:10.25781/KAUST-M5140

Relevant Courses: Applied Mathematics, Analog Integrated Circuits, Digital Design and Computer Architecture, Data Structures and Algorithms, Mathematical Foundations of Machine Learning, Digital Communication and Coding, and Principles of Estimation, Filtering, and Detection.

Double Major Bachelor of Science in Electrical Engineering & Physics

2013 - 2019

King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, KSA

GPA: 3.76/4.0 (1st Honors)

Exchange Student

Fall 2016

Georgia Institute of Technology, Atlanta, GA, USA

GPA: 3.5/4.0

Relevant Courses: Astronomy, Electromagnetism, Electrical Energy Conversion, and Signal Processing.

Summer School on Superconducting Electronics

Summer 2024

FLUXONICS - The European Foundry for Superconducting Electronics e.V., Corsica, France

Technical Skills

Core languages

- **Python**: scripting, tooling, verification and cocotb, data processing.
- **C++**: application and embedded development.
- **C**: foundation for embedded work, register-level code, and the C APIs that C++ touches.
- **Verilog / SystemVerilog**: RTL design and verification.

Supporting languages and tools

- **Tcl**: EDA tool automation (Vivado, Quartus, Synopsys, Cadence).
- **Make / CMake**: build systems for C, C++ and embedded projects.
- **Git**: version control, including branching, rebase and bisect.

Research Experience

Research & Development

Oct 2020 – Present

National Company of Telecommunications and Information Security (NTIS), KSA

- Developed direct RF sampling architectures with x16 DACs and x16 ADCs using an AMD/Xilinx RFSoc board.
- Developed and worked on different optical communication systems using an AMD/Xilinx MPSoC board.
- Developed PCB designs and FPGA-based (VHDL and Verilog) implementations for controllers and custom UART, I2C, and SPI drivers for embedded chips.

Master's Thesis Research

Spring 2023 – Spring 2024

Supervised by Dr. T. Al-Naffouri and Dr. M. Siala at KAUST, Thuwal, KSA

- Developed optimized Python scripts for efficient decoding algorithms to obtain BER plots and EXIT charts.
- Integrated FTN-inspired modulations with turbo equalization and LDPC decoding to improve the BER performance and achieve superior results to those reported in state-of-the-art literature.

Guided Research on Analog Electronic Circuits

Fall 2018 – Spring 2019

Supervised by Dr. M. Abuelma'atti at KFUPM, Dhahran, KSA

- Developed a memristor-based chaotic masking system built from off-the-shelf components and experimentally validated the synchronization of two chaotic oscillators.
- Developed a novel design of a tunable comb filter (based on CFOAs).

Mitacs Globalink Research Internship

Summer 2018

Supervised by Dr. S. Valluri at Western University, London ON, Canada

- Investigated the Lambert W function to describe the I-V characteristics of diode-based circuit models.

Visiting Student Research Program (VSRP) Internship

Summer 2017

Supervised by Dr. H. Fariborzi at KAUST, Thuwal, KSA

- Applied the Landau-Lifshitz-Gilbert (LLG) equations in Cadence using Verilog-A for All Spin Logic (ASL) circuits.
- Developed compact modules for charge/spin conduction, including ferromagnets (FM), non-magnets (NM), and FM/NM interfaces, along with generalizing the rotational matrix and optimizing the LLG solver.

Publications

1. M. Siala, A. Al-Nafisah, and T. Al-Naffouri, “Nyquist Signaling Modulation (NSM): An FTN-Inspired Paradigm Shift in Modulation Design for 6G and Beyond,” *arXiv:2511.08553*, 2025.
2. A. Al-Nafisah, D. Khan, S. Amara, and Y. Massoud, “Synchronization of Two Transistor-Based Chaotic Oscillators: Experimental Verification,” *IEEE 65th International Midwest Symposium on Circuits and Systems (MWSCAS)*, Fukuoka, Japan, pp. 1–4, 2022.
3. M. T. Abuelma’atti and A. Al-Nafisah, “New CFOA-Based Shadow Analog Comb Filter,” *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 1–8, 2020.
4. M. T. Abuelma’atti and A. Al-Nafisah, “Synchronization of Memristor-Based Chaotic Oscillator: Experimental Verification,” *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 21–28, 2020.
5. M. T. Abuelma’atti and A. Al-Nafisah, “A Memristor-Based Chaotic-Masking for Analog Spread-Spectrum Communication,” *Indonesian Journal of Electrical Engineering and Computer Science (IJECS)*, vol. 14, no. 2, pp. 966–971, 2019.

Posters

1. A. Al-Nafisah, M. Siala, and T. Al-Naffouri, “Utilizing EXIT Charts for Turbo Processing of M-ASK Modulations in 6G Communications,” *6G Summit*, Abu Dhabi, UAE, 2023.
2. A. Al-Nafisah, F. Al-Masri, F. Swaid, M. Al-Mousa, and M. Al-Shams, “Prototype of a Wireless Seismic Data Acquisition System,” *Senior Design Project, KFUPM*, Dhahran, KSA, 2019.
3. A. Al-Nafisah, N. Iqbal, and S. Al-Dharrab, “Throughput Analysis of Wireless Geophone Networks for Seismic Surveys,” *IEEE GCC Conference & Exhibition*, Bahrain, 2017.
4. A. Al-Nafisah, M. Alawein, and H. Fariborzi, “Modeling and Simulation of All-Spin Logic Device,” *KAUST Visiting Student Research Program (VSRP)*, Thuwal, KSA, 2017.